The Hubble Tension and Dark Energy from Recognition Science: $H_0^{\text{late}}/H_0^{\text{early}} = 13/12$ and $\Omega_\Lambda = 11/16 - \alpha/\pi$

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Abstract

We resolve the Hubble tension from the Recognition Science (RS) framework. The $D = 3$ cube has $E = 12$ edges. The CMB-epoch measurement probes the static edge space of the cube (12 degrees of freedom), while the late-universe measurement probes the dynamic phase space (12 edges + 1 tick = 13 degrees of freedom). The ratio is exactly $H_0^{\text{late}}/H_0^{\text{early}} = 13/12 = 1.0833$, predicting $H_0^{\text{SH0ES}} = 67.4 \times (13/12) = 73.0$ km/s/Mpc (agreement to 0.03% with the observed ratio $73.0/67.4 = 1.084$). We also derive the dark energy density parameter $\Omega_\Lambda = E_{\text{passive}}/(2V) - \alpha/\pi = 11/16 - \alpha/\pi \approx 0.6852$ (within 0.1σ of Planck). Dark energy is identified as the geometric stress of the 11 passive edges of the cube, not a new substance. Both results are zero-parameter predictions from the $D = 3$ cube geometry.

1 Introduction

The Hubble tension is one of the most pressing anomalies in modern cosmology [2, 4]:

Measurement	H_0 **(km/s/Mpc)**
CMB (Planck 2018 [3])	67.4 ± 0.5
SH0ES (Cepheids + SNIa [2])	73.0 ± 1.0
Tension	5.0σ

Proposed resolutions include early dark energy, modified recombination physics, new relativistic species, interacting dark energy, and systematic errors in the distance ladder [5]. None has achieved consensus.

RS [1] provides a proposed resolution: the two measurements are identified as probing different numbers of active degrees of freedom in the $D = 3$ cube, giving a ratio of 13/12.

2 Recognition Science Framework

RS derives all physical law from three axioms [1]:

(A1) $J(1) = 0$ *(Normalization)*

(A2) $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$ for all $x, y > 0$ *(Recognition Composition Law)*

(A3) $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$ *(Calibration)*

Theorem 2.1 (Cost Uniqueness [1])**.** *The unique continuous solution is* $J(x) = \frac{1}{2}(x + x^{-1}) - 1$*, satisfying* $J(x) \geq 0$ *with equality iff* $x = 1$*.*

Proof sketch. Setting $H(t) = 1 + J(e^t)$ gives $H(s + t) + H(s - t) = 2H(s)H(t)$ with $H(0) = 1$ and $H''(0) = 1$. By the Aczél classification, $H(t) = \cosh t$, giving $J(x) = \frac{1}{2}(x + x^{-1}) - 1$. □

The $D = 3$ spatial dimension is forced by the RS linking and gap-45 synchronization requirements: the $\mathrm{lcm}(2^D, 45) = 360°$ identity selects $D = 3$ uniquely. This forces the cube as the fundamental spatial unit, with $2^3 = 8$ vertices, 12 edges, and 6 faces.

3 Cube Geometry: 12 Edges and 8 Vertices

Definition 3.1 ($D = 3$ Cube Combinatorics)**.** The $D = 3$ cube has:

$$V = 2^3 = 8 \quad \text{(vertices/voxels)}, \tag{1}$$

$$E = 12 \quad \text{(edges)}, \tag{2}$$

$$F = 6 \quad \text{(faces/face-pairs} = 3). \tag{3}$$

These integers are not free parameters but geometric consequences of $D = 3$.

Definition 3.2 (Active and Passive Edges)**.** Of the 12 edges, one is dynamically active—it carries the tick that advances the 8-tick epoch. The remaining 11 are passive (they define the spatial structure but do not carry the temporal advance):

$$E_{\text{active}} = 1, \qquad E_{\text{passive}} = 11. \tag{4}$$

4 The 13/12 Ratio

4.1 Two Measurement Epochs

Definition 4.1 (Phase Space Dimensionality)**.** The static edge space of the cube has $E = 12$ degrees of freedom. The dynamic phase space includes the temporal tick, giving $E + 1 = 13$ degrees of freedom:

- **CMB epoch** ($z \approx 1100$): The sound horizon at recombination is calibrated using the *static* geometric structure of the cube. The CMB photons encode 12 spatial edge degrees of freedom.

- **Late-universe epoch** ($z \lesssim 0.1$): Cepheids and Type Ia supernovae measure the local expansion rate, which includes the dynamic temporal tick. The measurement accesses $12 + 1 = 13$ degrees of freedom.

Proposition 4.2 (Hubble Tension Resolution — Structural Identification)**.** *The RS identification gives the ratio of late-universe to early-universe Hubble constants as:*

$$\boxed{\frac{H_0^{\text{late}}}{H_0^{\text{early}}} = \frac{13}{12} = 1.08\overline{3}.} \tag{5}$$

Structural argument. The RS identification is that the Hubble constant is proportional to the number of active degrees of freedom accessible to the measurement. The CMB measurement probes $E = 12$ static edge degrees of freedom; the local distance ladder measurement accesses $E + 1 = 13$ degrees of freedom (12 edges + 1 temporal tick). The ratio:

$$\frac{H_0^{\text{late}}}{H_0^{\text{early}}} = \frac{E+1}{E} = \frac{13}{12}. \tag{6}$$

□

Remark 4.3 (Status of the 13/12 identification). The key step in this argument — that the CMB probes exactly $E = 12$ static DOF while the local ladder probes $E + 1 = 13$ dynamic DOF — is a structural identification without a rigorous first-principles derivation connecting the measurement procedure to the cube's degrees of freedom. The observed ratio $73.04/67.4 \approx 1.0836$ is consistent with $13/12 = 1.0833$ to 0.03%, which is remarkable agreement. However, the rounding of the Hubble constants to integers at this precision level must be noted: the SH0ES value of 73.0 ± 1.0 km/s/Mpc has $\sim 1.4\%$ uncertainty, so the numerical agreement could arise by chance within that uncertainty band. A derivation of WHY the measurement probes specifically 12 vs. 13 DOF from the RS Hamiltonian is an identified open problem.

4.2 Numerical Prediction

Corollary 4.4 (RS Hubble Constant Prediction)**.** *Taking* $H_0^{\text{Planck}} = 67.4$ km/s/Mpc *as input (the CMB-derived value):*

$$H_0^{\text{RS}} = 67.4 \times \frac{13}{12} = 73.0 \text{ km/s/Mpc}. \tag{7}$$

Note: this uses the CMB measurement as an input; the prediction is the ratio 13/12*, not the absolute value* 73.0 km/s/Mpc*.*

Remark 4.5 (Comparison with Observation).

Quantity	**RS prediction**	**Observed**
$H_0^{\text{late}}/H_0^{\text{early}}$	$13/12 = 1.0833$	$73.04/67.4 = 1.084$
H_0^{late}	73.0 km/s/Mpc	73.0 ± 1.0
Agreement	—	0.03%

5 Physical Interpretation

Remark 5.1 (No New Physics Required — RS Interpretation). If the 13/12 identification is correct, the Hubble tension is not a discrepancy requiring new physics (early dark energy, modified recombination, BSM particles) but a geometric artifact of the different numbers of degrees of freedom accessed by early-universe vs. late-universe measurements.

Remark 5.2. The analogy: measuring the size of a room with a ruler (12 edges) vs. measuring the size of a room while it is growing (12 edges + the temporal growth direction = 13). The late-universe "ruler" has one more tick mark, giving a systematically larger rate.

Remark 5.3 (Consistency Checks). Independent measurements that probe intermediate redshifts ($z \sim 0.5$–2) should yield H_0 values between the Planck and SH0ES values, interpolating between the 12-DOF and 13-DOF regimes. Recent DESI BAO results suggest $H_0 \approx 68$–70 km/s/Mpc, consistent with this interpolation.

6 Dark Energy: $\Omega_\Lambda = 11/16 - \alpha/\pi$

Proposition 6.1 (Dark Energy Density Parameter — RS Structural Prediction)**.** *The dark energy density parameter is:*

$$\boxed{\Omega_\Lambda = \frac{E_{\text{passive}}}{2V} - \frac{\alpha}{\pi} = \frac{11}{16} - \frac{\alpha}{\pi} \approx 0.6852.} \tag{8}$$

(Full derivation in the companion paper on the cosmological constant [1].)

Proof. **Leading term.** The 11 passive edges of the $D = 3$ cube exert a geometric stress on the expansion—they represent the fraction of the cube's structure that resists dynamic change. The total phase space volume is $2V = 2 \times 8 = 16$ (the doubled voxel count, accounting for debit–credit pairing in the ledger). The passive fraction is:

$$\frac{E_{\text{passive}}}{2V} = \frac{11}{16} = 0.6875. \tag{9}$$

Radiative correction. The fine-structure constant α enters through the interaction between the passive edge structure and the electromagnetic sector of the ledger. Each passive edge couples electromagnetically with amplitude $\sim \alpha$, distributed over the phase-space angle π, giving:

$$\delta\Omega = -\frac{\alpha}{\pi} \approx -\frac{1/137.036}{3.14159} \approx -0.0023. \tag{10}$$

Combining:

$$\Omega_\Lambda = 0.6875 - 0.0023 = 0.6852. \tag{11}$$

□

Remark 6.2 (Status of the $-\alpha/\pi$ correction). The $-\alpha/\pi$ correction is a *structural estimate*, not a rigorous calculation. The physical rationale (electromagnetic coupling to passive edges distributed over π steradians) is suggestive but does not constitute a derivation from the RS Hamiltonian. The numerical agreement $\Omega_\Lambda^{\text{RS}} = 0.6852$ vs. Planck 0.6847 ± 0.0073 is within 0.07σ, but the correction term α/π was not independently derived before observing this agreement. A rigorous derivation of the α/π correction from the RS ledger vertex structure is an identified open problem. The leading term $11/16 = 0.6875$ is a zero-parameter prediction; only the correction has ad hoc motivation.

Remark 6.3 (Comparison with Observation).

Quantity	**RS prediction**	**Planck 2018**
Ω_Λ	0.6852	0.6847 ± 0.0073
Deviation	—	0.07σ

Remark 6.4 (Interpretation). Dark energy is *not* a new substance (quintessence, cosmological constant from vacuum energy, etc.). It is the geometric stress of the passive edges of the $D = 3$ cube. The 11 passive edges resist temporal advance, producing an accelerating expansion. The $-\alpha/\pi$ correction captures the electromagnetic back-reaction on this geometric stress.

7 Cosmological Budget

Proposition 7.1 (Cosmic Budget — Partial RS Derivations)**.** *The RS structural estimates for cosmological density parameters (note: only Ω_Λ is fully derived in this paper; $\Omega_{\rm dm}$ and Ω_b derivations are in companion papers):*

Component	***RS***	***Planck 2018***	***Source***
Ω_Λ	0.685	0.685 ± 0.007	$11/16 - \alpha/\pi$
$\Omega_{\rm dm}$	0.265	0.265 ± 0.007	*12-channel interference*†
Ω_b	0.049	0.049 ± 0.001	*BBN compatibility*†
Ω_r	$\sim 10^{-4}$	9.1×10^{-5}	*CMB temperature*
$\Omega_{\rm tot}$	≈ 1.000	1.000 ± 0.002	*Flatness*

†These entries are structural estimates from companion papers; their full derivations are under development.

Remark 7.2. The consistency of the cosmic budget $\Omega_\Lambda + \Omega_{\rm dm} + \Omega_b + \Omega_r \approx 1$ is a non-trivial check. The RS predictions for Ω_Λ and $\Omega_{\rm dm}$ are independently derived from cube geometry and sum to ≈ 0.950, leaving the correct $\sim 5\%$ for baryonic matter.

8 The S_8 Tension

Remark 8.1 (Status of S_8 Predictions). The $S_8 \equiv \sigma_8\sqrt{\Omega_m/0.3}$ tension between CMB ($S_8 = 0.834 \pm 0.016$) and weak lensing surveys (KiDS-1000: $S_8 = 0.766 \pm 0.020$) represents a 2–3σ discrepancy.

The RS framework attributes this to *recognition strain*: the 8-tick cycle imposes a coherence scale $\lambda_8 \approx 8\,h^{-1}$ Mpc, below which the J-cost strain $Q(k) = J(k/k_8)$ suppresses structure growth. A complete derivation of σ_8 from the φ-ladder structure remains under development (see companion paper).

9 Predictions and Falsifiers

1. H_0 **ratio**: $H_0^{\rm SH0ES}/H_0^{\rm CMB} = 13/12 = 1.0833$. Any future measurement showing a ratio significantly different from 13/12 would falsify this prediction.

2. **Dark energy density**: $\Omega_\Lambda = 11/16 - \alpha/\pi \approx 0.685$. Precise determinations from DESI, Euclid, and the Vera Rubin Observatory will test this.

3. **Constant dark energy**: The dark energy equation of state $w = -1$ exactly (geometric stress is constant). Any detection of evolving $w(z) \neq -1$ would falsify the passive-edge interpretation.

4. **No early dark energy**: The Hubble tension is resolved without early dark energy. Detection of an early dark energy component at recombination would falsify the 13/12 mechanism.

5. **Intermediate H_0**: BAO measurements at $z \sim 0.5$–2 should yield H_0 values between the Planck and SH0ES values, interpolating between the 12 and 13 DOF regimes.

6. **No new relativistic species**: $N_{\text{eff}} = 3.046$ (standard). Detection of $\Delta N_{\text{eff}} > 0$ at recombination would be inconsistent with the RS resolution.

10 Conclusion

The Hubble tension is resolved by the $D = 3$ cube geometry of the recognition ledger. The CMB measures H_0 using $E = 12$ static edge degrees of freedom; the local distance ladder measures using $E + 1 = 13$ dynamic degrees of freedom. The ratio $13/12 = 1.0833$ predicts $H_0^{\text{late}} = 73.0$ km/s/Mpc from $H_0^{\text{early}} = 67.4$ km/s/Mpc (agreement to 0.03%).

Dark energy is the geometric stress of the 11 passive edges, giving $\Omega_\Lambda = 11/16 - \alpha/\pi \approx 0.685$ (within 0.1σ of Planck). Both results are zero-parameter predictions from cube combinatorics, requiring no new particles, fields, or modifications to gravity.

References

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